

A theater owner is looking to maximize his profits and needs to increase his ticket prices. The function $P(t) = -50t^2 + 100t + 14,400$ represents his total profit, P , based on the amount, in dollars, he raises the ticket price, t .

1. Rewrite the function in vertex form.

2. Rewrite the function in factored form.

Complete the statements.

3. The ☐ vertex
☐ factored
☐ standard form of the function can be used to determine the interval on which the theater owner experiences a profit. The theater owner experiences a profit on the interval _____.

4. The ☐ vertex
☐ factored
☐ standard form of the function can be used to determine when the theater owner experiences the greatest profit. The theater owner experiences the greatest profit when _____.

5. The ☐ vertex
☐ factored
☐ standard form of the function can be used to determine when the theater owner experiences neither a profit nor a loss. This occurs when _____.

The function $H(x) = 0.5x^2 - x + 33$ models the amount of people, in millions, P , enrolled in an HMO insurance in the United States in the years, x , since 1990.

6. What would the zeros of this function reveal in this context?

7. Rewrite the function in vertex form.

Complete the statements.

8. The

☐ vertex

☐ factored

☐ standard

 form of the function can be used to determine the year with the lowest HMO enrollment.

9. The year with the lowest HMO enrollment was _____ when _____ million people were enrolled.

10. The

☐ vertex

☐ factored

☐ standard

 form of the function can be used to determine how many people, in millions, were enrolled in an HMO in 1990, _____ million people.